

The quantum twisting microscope

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The invention of scanning probe microscopy revolutionized the way electronic phenomena are visualized¹. Whereas present-day probes can access a variety of electronic properties at a single location in space², a scanning microscope that can directly probe the quantum mechanical existence of an electron at several locations would provide direct access to key quantum properties of electronic systems, so far unreachable. Here, we demonstrate a conceptually new type of scanning probe microscope—the quantum twisting microscope (QTM)—capable of performing local interference experiments at its tip. The QTM is based on a unique van der Waals tip, allowing the creation of pristine two-dimensional junctions, which provide a multitude of coherently interfering paths for an electron to tunnel into a sample. With the addition of a continuously scanned twist angle between the tip and sample, this microscope probes electrons along a line in momentum space similar to how a scanning tunnelling microscope probes electrons along a line in real space. Through a series of experiments, we demonstrate room-temperature quantum coherence at the tip, study the twist angle evolution of twisted bilayer graphene, directly image the energy bands of monolayer and twisted bilayer graphene and, finally, apply large local pressures while visualizing the gradual flattening of the low-energy band of twisted bilayer graphene. The QTM opens the way for new classes of experiments on quantum materials.

An electron in a solid is a quantum mechanical object best described by an extended wavefunction, reflecting its existence in a superposition of spatial locations. The invention of the scanning tunnelling microscope (STM)¹ dramatically changed how we can visualize electrons in real space. It was followed by a large array of other scanning probe techniques that now allow local measurements of various electronic properties². So far, existing scanning microscopes can only probe electronic properties at one location at a time. Therefore, they are unable to probe the interference of several tunnelling trajectories, which would map the evolution of the quantum mechanical phase in space. To do so requires a scanning interferometer that probes an electron at several locations simultaneously and quantum coherently.

In this work, we demonstrate a conceptually new type of scanning probe microscope—the quantum twisting microscope (QTM)—capable of performing local interference experiments at its tip. Our technique is based on a unique van der Waals (vdW) tip, which is brought into contact with a vdW sample, forming a two-dimensional (2D) interface that can be twisted and spatially scanned with high angular and positional precision. The QTM enables two orthogonal and complementary classes of experiments. (1) ‘In situ twistronics’, measuring a continuously twistable interface between a pair of vdW materials (for example, twisted bilayer graphene, TBG^{3–5}). Here, we place the active vdW layers in the tip and sample in direct contact, such that their wavefunctions strongly couple, and probe the transport properties of the hybridized interface. (2) Momentum-resolved tunnelling—here we insert a tunnel barrier between the active vdW layers in the tip and sample. This barrier decouples their wavefunctions, allowing the tip to act as a non-invasive momentum-resolved probe of the sample’s energy bands. As an electron can coherently tunnel into a sample at many

locations along the 2D interface, this junction acts as an interferometer on a tip. Specifically, in the absence of scattering, an electron would tunnel only between states of equal momenta^{6–16}. In this modality, the microscope’s twisting is used for scanning an arc in the momentum space of the sample, and imaging along it the sample’s energy bands.

The QTM’s operating principle

So far, existing in situ twisting experiments have been based on devices with a rotatable part that is pushed by an external mechanical device (for example, atomic force microscope (AFM) tip)^{17–21}. Our QTM, by contrast, elevates the AFM tip to become an integral part of the twisted device, which is now split into two parts: a standard vdW device on a flat substrate (Fig. 1a and Supplementary Information Section 2) and a vdW device formed on a specially designed pyramid at the edge of an AFM cantilever (Fig. 1b and Supplementary Information Section 1). Both sides have independent electrical contacts. We use a commercial AFM to bring the two parts into contact and maintain a constant force across the interface throughout the entire experiment (Fig. 1c). On the AFM stage, we mount a piezoelectric rotator with X and Y nanopositioners on top (Supplementary Information Section 3). This setup allows rotating the bottom sample with 0.001° resolution and positioning the sample’s point-of-interest at the centre of rotation.

A crucial ingredient of the QTM is its tip design, facilitating the formation of a flat vdW plateau at its apex. To achieve this, we start with focused ion beam (FIB) deposition of a platinum pyramid (roughly 1.2–1.6 µm tall) on a tipless AFM cantilever (Fig. 1e). This is followed by sequentially transferring graphite, hBN and the active vdW layer (for

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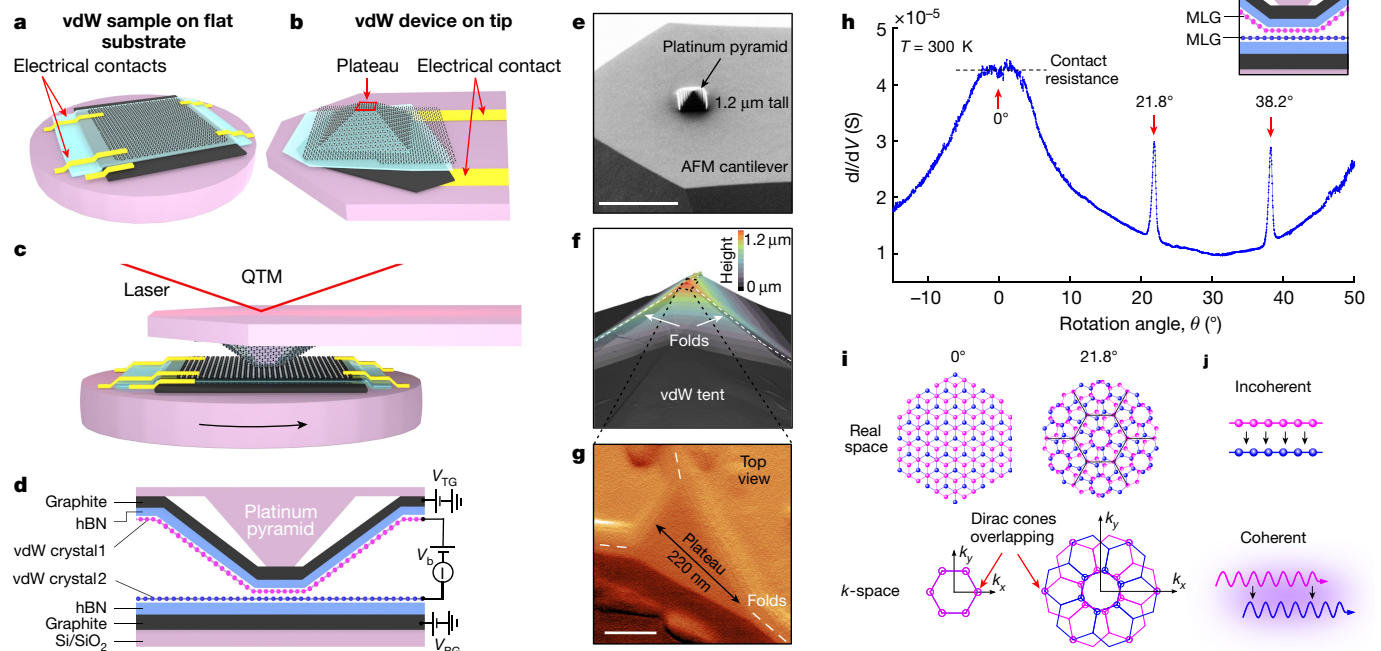


Fig. 1 | The QTM setup and in situ twistronics experiments. **a–c**, Illustrations of the components of the QTM. **a**, vdW heterostructure assembled on a flat substrate with electrical contacts. **b**, vdW heterostructure assembled on a pyramid positioned near the edge of an AFM cantilever, with electrical contacts. At the apex of the pyramid, the vdW device-on-tip has a flat plateau. **c**, These devices are brought into contact using a commercial AFM, fitted with a piezoelectric rotator that allows to control the relative angle between the tip and sample, θ , continuously with a 0.001° resolution, in addition to the usual lateral (X or Y) scanning capabilities of the AFM. **d**, Measurement circuit plotted over a schematic cross-section of the QTM junction. A voltage bias, V_b , is applied between the two active vdW layers (vdW crystals 1 and 2), and the corresponding current, I , is measured. Buried top and bottom graphite gate voltages, V_{BG} and V_{TG} , can modify the local carrier density and electric field in the junction (in this paper, the top gate is physically shorted to the top graphene layer such that V_{TG} is identically zero). **e**, SEM image of an AFM cantilever with a custom-made

platinum pyramid deposited by a FIB. Scale bar, $5 \mu\text{m}$. **f**, The topography of a vdW device-on-tip (graphite/hBN/MLG), imaged by AFM. The vdW layers form a tent over the Pt pyramid, with folds (dashed white) leading to a flat plateau. **g**, Zoomed-in AFM image around the pyramid's apex (peak-force-error signal), showing the spontaneously formed flat plateau. Scale bar, 100 nm . **h**, Measured conductance, dI/dV , versus rotation angle, θ , between two graphene monolayers (MLG, top inset) in direct contact, $V_b = 50 \text{ mV}$, $V_{BG} = 0$, $T = 300 \text{ K}$. The two vdW devices are kept in continuous contact throughout the measurement. **i**, Real-space and momentum-space registry for the commensurate angles of 0° and 21.8° . **j**, Illustrations of the incoherent and coherent tunnelling across the 2D junction. In the former, electrons tunnelling at various locations are incoherent and the tunnelling is proportional only to the local wavefunction squared. In the latter, tunnelling trajectories within the coherence length interfere and the tunnelling is sensitive also to the variation of the phases of the top and bottom wavefunctions along the junction.

example, monolayer graphene, MLG) on the pyramid using a polymer membrane²². The graphite screens the substrate's disorder potential, and the hBN acts as a spacer. An AFM image of the resulting tip (Fig. 1f) shows that the vdW stack forms a 'tent' over the pyramid with folds climbing up its apex. At the apex, a flat plateau spontaneously forms, whose corners are determined by the folds (Fig. 1g). Apart from a small touching point at the apex, the tent is mostly suspended (Supplementary Information Section 1). The formed plateau is wider than the pyramid apex, resting on it as a pivoting point. Therefore, on contact with the sample, the plateau self-aligns to become parallel to the sample. By varying the flake thicknesses we achieve plateaus of varying linear dimensions between 50 nm and $1 \mu\text{m}$. These dimensions are small enough to make our QTM tip a local scanning probe, yet large enough (roughly $1,000$ atoms across or more) for wavefunctions on the plateau to have a well-defined momentum ($\Delta P \approx 1/1,000$ of the Brillouin zone size). Experiments involve a bias voltage, V_b , applied between the two active layers (Fig. 1d) and measurement of the corresponding current, I . Additionally, buried bottom and top graphite layers can serve as gates for the junction.

In situ twistronics

We start with a 'twistronic' measurement of a twistable interface between two graphene monolayers (MLG) in direct contact. This interface has remained elusive for existing in situ twistronics techniques^{17–19} because

MLG easily crumbles on twisting. Figure 1h shows the MLG–MLG interface conductance, dI/dV , measured versus twist angle, θ , at $T = 300 \text{ K}$. The sample and tip are continuously in contact throughout the measurement. The layers do not show signs of locking at $\theta = 0^\circ$, and the conductance traces are highly reproducible, highlighting the mechanical robustness of the QTM junction. The conductance is mirror symmetric around $\theta = 30^\circ$, ($\theta \rightarrow 60^\circ - \theta$), where its value is minimal. It rises continuously towards $\theta = 0^\circ$ but plateaus at small angles ($|\theta| \lesssim 4^\circ$) limited by the resistance away from the tip ('contact resistance'). Similar to the measurements in graphite–graphite¹⁷ and graphite–MLG¹⁸ interfaces, we see sharp conductance peaks at $\theta = 21.8^\circ$ and 38.2° . At these angles, the two layers form commensurate stackings in real space^{17,18,23,24} with a $\sqrt{7} \times \sqrt{7}$ supercell (Fig. 1i).

The conductance peaks may reflect real-space registry of the atoms in the two layers. However, by definition, whenever unit cells are commensurate in real space, their corresponding Brillouin zones are also commensurate in k -space, indicating that momentum states are also matched. Specifically, at $\theta = 0^\circ$ the Dirac cones of the two layers overlap at the corners of the first Brillouin zone, and at $\theta = 21.8^\circ$ they overlap at the corners of the third Brillouin zone (Fig. 1i). The relevance of real versus momentum-space matching is intimately connected to the quantum coherence of the 2D tunnel junction: in an incoherent junction, tunnelling events at different locations are independent, summing up classically to yield the total current (Fig. 1j, top). In such a case, only the real-space matching is relevant. Conversely, in a coherent junction, tunnelling events at different locations interfere, yielding a current

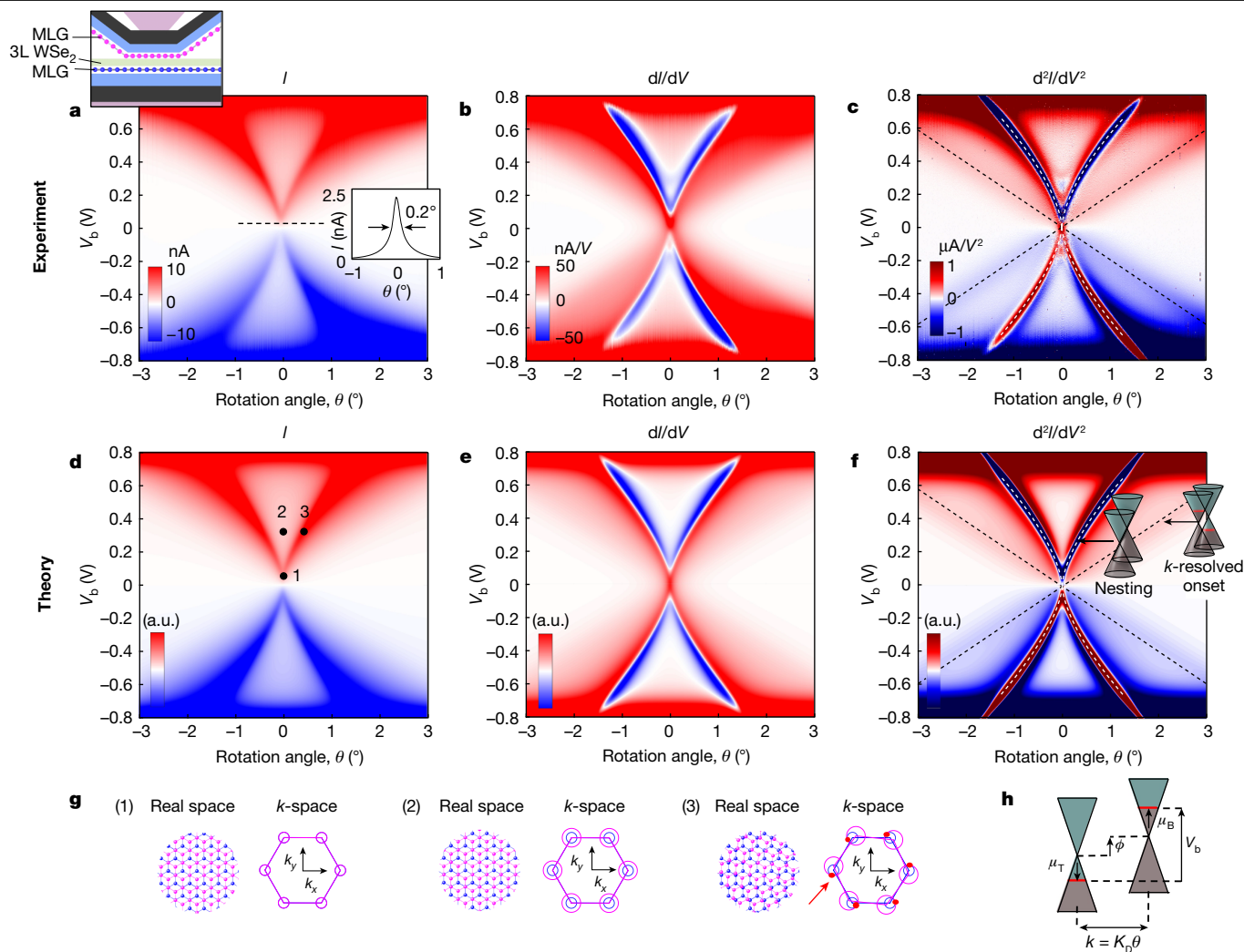


Fig. 2 | Momentum-resolved tunnelling between two twisted graphene monolayers. **a**, Top inset shows a schematic cross-section of the experiment, comprising an MLG/trilayer WSe₂/MLG tunnel junction. The WSe₂ tunnel barrier suppresses the hybridization between the MLG layers allowing one to probe the unperturbed energy bands of the other. The main panel shows the tunnelling current, I , measured as a function of rotation angle, θ , and interlayer bias, V_b , at $T = 300$ K. Right inset shows a plot of I versus θ at $V_b = 40$ mV (along the black dashed line in the main panel). The peak's full-width at half-maximum is 0.2° . **b**, A lock-in measurement of the differential conductance, dI/dV , as a function of θ and V_b , showing strong negative differential resistance (NDR, blue) along a curved X feature. The measurement shows a good symmetry between the positive and negative bias directions. **c**, The second derivative, d^2I/dV^2 , versus θ and V_b , obtained by a numerical derivative of the data in **b**. The black and white dashed lines correspond to two specific alignment conditions between the bands, 'k-resolved onset' and 'nesting', shown in **f** and described in the main text and Supplementary Information Section 5 and Supplementary

Video 4. **d–f**, Theoretically calculated momentum-conserving tunnelling I , dI/dV and d^2I/dV^2 between two MLG layers spaced by a tunnelling barrier, as a function of θ and V_b , based on the Bistritzer–MacDonald expression for momentum-conserving tunnelling between rotated vdW layers^{23,27}. The calculation includes finite temperature, $k_B T = 25$ meV, and a finite inverse electron lifetime¹², $\gamma = \gamma_0 + \gamma_1 |\epsilon - \mu|$, where ϵ is the electron energy and μ is the chemical potential. $\gamma_0 = 4$ meV, $\gamma_1 = 0.035$ are the experimentally fitted values. The illustrations show the relative alignment of the two Dirac cones at the 'k-resolved onset' and 'nesting' conditions, along the corresponding dashed black and dashed white lines. **g**, Schematics 1–3 illustrate the relative alignment of the two MLG layers in real space and momentum space, corresponding to points 1–3 in **d**. **h**, Schematic band alignment of the two layers under finite V_b and θ . V_b divides between shifts to the chemical potentials of bottom and top layers, μ_b and μ_t , and the energy band shift (equal to the electrostatic potential shift), ϕ . The Dirac points are shifted in momentum by $k = K_D \theta$, where K_D is the momentum of the Dirac point. a.u., arbitrary units.

that is sensitive also to the local phases of the wavefunctions (Fig. 1j, bottom). In particular, tunnelling is possible only between wavefunctions with matching energy and momentum.

Local momentum-resolved tunnelling

To observe the momentum-conserving nature of the QTM we add a tunnelling barrier between the two MLG layers (trilayer WSe₂). This barrier suppresses the hybridization between tip and sample, enabling the tip to act as a probe for the unperturbed energy bands of the sample. The barrier also notably increases the tunnel junction resistance,

assuring that an applied bias falls predominantly across this junction and that the measurement is not affected by contact resistance even near $\theta = 0^\circ$. Figure 2a shows the measured tunnelling current, I , versus the interlayer bias, V_b , and θ at $T = 300$ K. Around $\theta = 0^\circ$, I increases slowly with V_b at low bias and then sharply increases along a curved X feature in the θ – V_b plane. This increase is followed by a sharp drop at slightly higher V_b . At much higher biases (roughly 0.8 V), I rises again, this time exponentially with V_b , and rather homogeneously for all θ . In the measured conductance, dI/dV (Fig. 2b), the sharp drop of I manifests as a strong negative differential resistance^{9–12,14,16}. Finer details appear in the second derivative, d^2I/dV^2 (Fig. 2c): in addition to the strong curved X (dashed

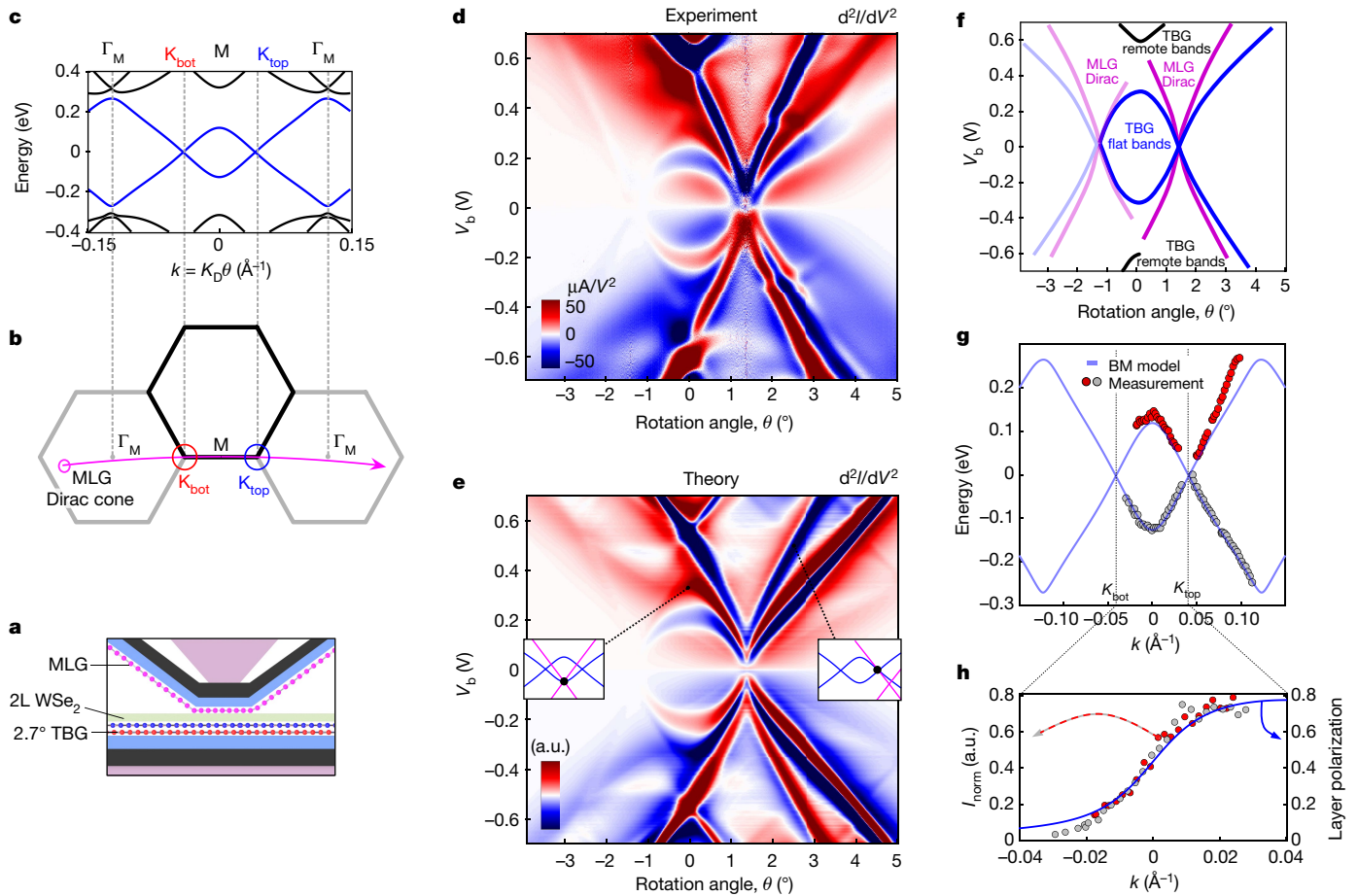


Fig. 3 | QTM imaging of the energy bands of TBG. **a**, Schematic of the experiment, composed of a MLG/bilayer WSe_2 /2.7° TBG tunnel junction. **b**, The mini-Brillouin zone of the TBG, with the Dirac cones of the underlying top and bottom layers (represented by blue and red circles) located at two K -points (K_{bot} and K_{top}). Also shown are the Γ_M points of adjacent mini-Brillouin zones. When the MLG probe layer is rotated with respect to the TBG, its Dirac cone (pink circle) traces the TBG energy bands along an arc in momentum space (spink), crossing through the K_{bot} and K_{top} points, and very close to the Γ_M points. **c**, Theoretical energy bands of 2.7° TBG along the pink arc in **b**, calculated using the Bistritzer–MacDonald model³. The flat and remote bands are shown in blue and black, respectively. **d**, The second derivative, d^2I/dV^2 , measured as a function of θ and V_b at $T = 300$ K. **e**, Theoretically calculated d^2I/dV^2 for this junction, as a function of θ and V_b . The theory is based on the Bistritzer–MacDonald expression for momentum-conserving tunnelling^{3,23,27} and includes finite temperature, $k_B T = 25$ meV, and finite inverse electron lifetime¹², $\gamma = \gamma_0 + \gamma_1 |\epsilon - \mu|$

with $\gamma_0 = 4$ meV, $\gamma_1 = 0.02$. The insets show the relative alignment of the MLG and TBG energy bands at two points in the θ – V_b plane. The left inset shows an instance where the MLG Dirac point matches in energy and momentum a point on the flat TBG bands. The right inset shows an instance where the TBG Dirac point matches in energy and momentum a point on the MLG Dirac band. **f**, Tracing of the main features of the measurement in **d**, including features related to the TBG flat (blue) and remote (black) bands, as well as two copies of the MLG Dirac bands (purple). **g**, The TBG energy versus momentum dispersion (red and grey dots) determined from the measurement in **d** after converting V_b to the energy axis (main text), together with the bands of the Bistritzer–MacDonald (BM) model (blue). **h**, The magnitude of the tunnelling current, I , traced along the features that correspond to the flat bands (blue curves, **f**) between K_{bot} and K_{top} (grey/red points for the positive/negative bias side). We normalize the current values by a constant to compare with the theory. Blue line denotes the theoretically calculated layer polarization of the wavefunction versus momentum.

white lines), we observe a straight X feature (dashed black lines), along which d^2I/dV^2 shows peaks (dips) on the positive (negative) bias side.

To understand the curved X feature, consider three points in the θ – V_b plane (Fig. 2d): at point 1 ($\theta = 0^\circ$, small V_b) the atoms of the two layers are registered in real space and their energy surfaces are matched in momentum space (Fig. 2g, panel 1). Increasing V_b while keeping $\theta = 0^\circ$ (point 2) maintains the real-space registry but offsets the relative energies of the Dirac cones. Consequently, for almost all energy slices within the bias window, the equal energy states have mismatched momenta (one slice is shown in Fig. 2g, panel 2). The observed drop of I is therefore concomitant with the loss of energy-momentum-matched eigenstates in the two layers. By twisting to a finite θ while maintaining V_b (point 3), the layers lose the real-space registry but regain overlap of states in momentum space, enabling momentum-conserving tunnelling processes (Fig. 2g panel 3). Indeed, at this point I becomes large again, demonstrating its momentum-conserving and quantum

coherent nature (Supplementary Information Section 13). The extremely narrow peak of I versus θ ($\Delta\theta \approx 0.2^\circ$, Fig. 2a, right inset) indicates an excellent momentum resolution, roughly 0.004 of the Brillouin zone size, comparable to state-of-the-art angle-resolved photoemission spectroscopy (ARPES) detectors^{25,26}. Additional experiments with finite back gate voltages (Supplementary Information Section 6) further show a rich evolution of the features with the total charge in the system.

Theoretical calculation of momentum-resolved tunnelling between twisted layers^{23,27} (Fig. 2d–f and Supplementary Information Section 4), shows excellent agreement with the experiments and identifies the observed features: the straight X reflects the onset of momentum-resolved tunnelling, happening when the Fermi surface of one layer touches the empty bands of the other layer (k -resolved onset, Fig. 2f). The curved X corresponds to nesting of the energy bands (nesting, Fig. 2f), with many energy-momentum-conserving states available to tunnel, explaining the large increase of I along this feature (Supplementary Video 4). With

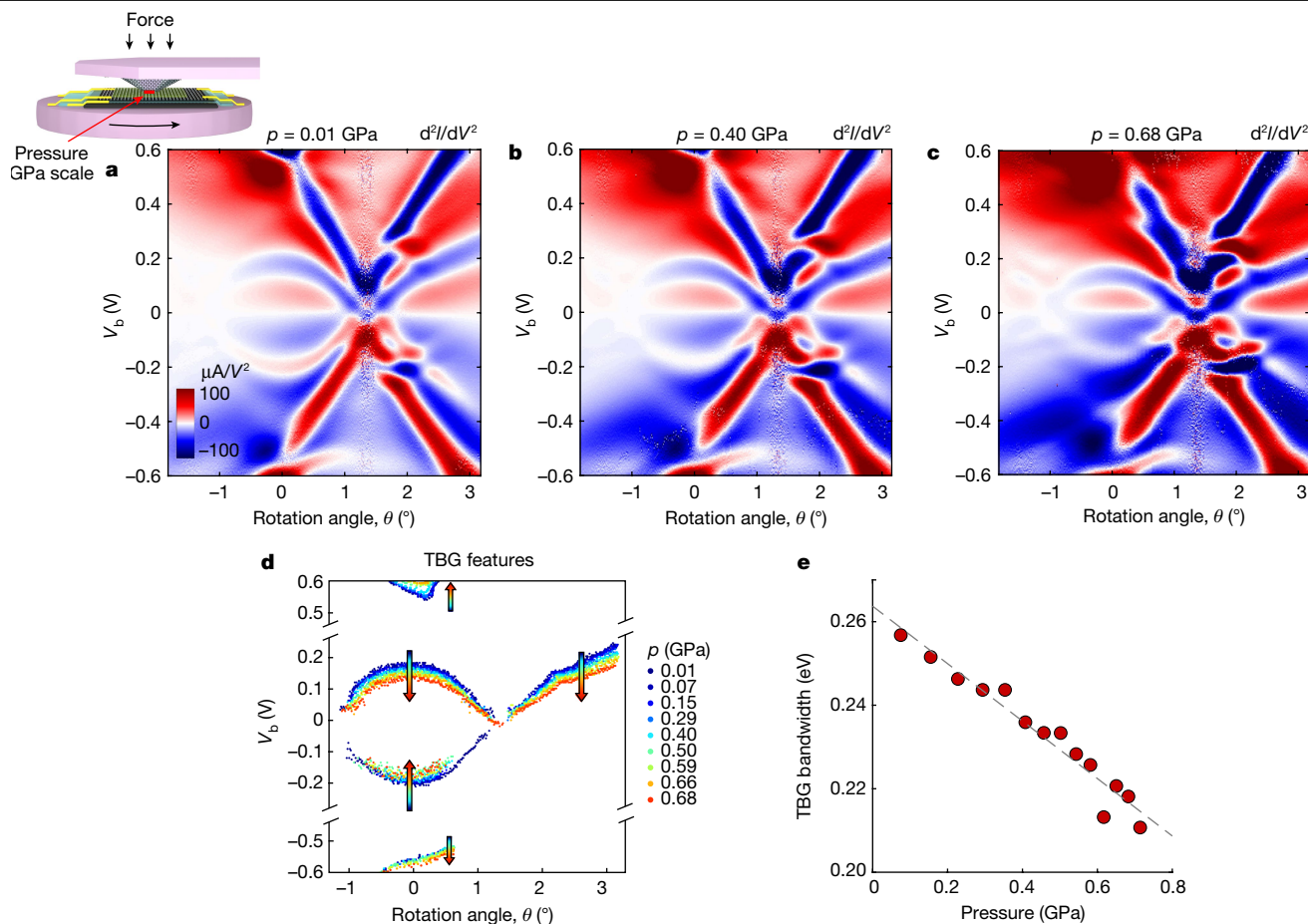


Fig. 4 | Imaging the effects of applied pressure on the TBG energy bands.

a, The inset shows an illustration of the experiment: applying a fixed AFM force across the junction results in GPa-scale pressures at the interface. In the experiment, we keep the pressure on while twisting. **a–c**, Measured d^2I/dV^2 versus θ and V_b for the junction in Fig. 3, but under applied pressures of $p = 0.01$, $p = 0.4$ and $p = 0.68$ GPa. $T = 300$ K. **d**, Evolution of the TBG energy bands with p . In this figure, we trace features that reflect the flat and remote bands as a function of pressure (key). For every pressure, we plot points that were obtained from tracing the relevant feature ('onset' feature for the flat bands, zero dI^2/dV^2

contour for the remote bands). Visibly, with increasing p , the flat bands gradually shrink towards zero V_b while the remote bands shift to higher V_b (arrows), consistent with a pressure-controlled anticrossing. **e**, The bandwidth at the M point of the TBG flat energy bands versus p (red dots), determined from linecuts at $\theta = 0^\circ$. We convert V_b to the energy from the simultaneously measured MLG Dirac bands (curved X on the right side in **a–c**, Supplementary Information Section 8). Dashed line is a linear fit to the data. At the highest applied pressure (0.68 GPa), the bandwidth shrinks by roughly 17%.

a further increase of V_b or a further decrease of θ most states cease to be energy-momentum conserving, explaining the consequent drop in I .

For MLG, the onset condition appears at $V_b = \hbar v_F K_D \theta$, where K_D is the Dirac point momentum (Supplementary Information Section 5), namely, V_b and θ substitute energy and momentum in the Dirac equation. This fits excellently the measurement with $v_F = 1.05 \pm 0.02 \times 10^6$ m s⁻¹ (dashed black lines, Fig. 2c), consistent with measured graphene Fermi velocity²⁸. Additionally, the nesting line provides the energy shift between the bands, ϕ , at any value of V_b , from which the electronic compressibility of the system can be straightforwardly determined (Supplementary Information Section 5).

Momentum-resolved imaging of TBG energy bands

Having probed the energy bands of MLG, we now turn to TBG. The experiment consists of MLG probe, bilayer WSe₂ barrier and TBG with a 2.7° twist (Fig. 3a). In momentum space, the TBG mini-Brillouin zone hosts at its corners the Dirac cones of its top and bottom graphene sheets (red and blue circles at K_{top} and K_{bot} , Fig. 3b). When the MLG is rotated, its Dirac cone (pink circle, Fig. 3b) scans the TBG's energy bands along a constant-radius arc in momentum space (pink, Fig. 3b), cutting precisely through K_{top} and K_{bot} and passing close to the Γ_M points (centre points) of

adjacent mini-Brillouin zones. Along this arc, the TBG is theoretically predicted to show 'flat' bands around zero energy (note that by 'flat', we mean that at a twist angle of 2.7°, the dispersion is less than graphene's Dirac dispersion but not entirely flat) and remote bands at higher energies^{3,29} (Fig. 3c). Figure 3d shows the dI^2/dV^2 measured versus V_b and θ . The second derivative diminishes the smoothly evolving background, revealing key features (traced in Fig. 3f). It shows a superposition of the 'system' and 'probe' energy bands, including the TBG's flat bands (blue), remote bands (black) and two copies of the MLG Dirac bands (purple).

A theoretical calculation of the MLG–TBG momentum-resolved tunnelling (Fig. 3e) agrees well with the experiment. Specifically, it reproduces the flat and remote TBG band features, and the two displaced copies of the MLG Dirac bands. It further shows that the band imaging is effectively performed by Dirac points: for example, when the MLG Dirac point matches in energy and momentum a state on the TBG flat bands (left inset, Fig. 3e), the dI^2/dV^2 shows a peak. This peak corresponds to a minimum in I , reflecting the minimal tunnelling density of states at the Dirac point. Similarly, when one of the two TBG's Dirac points matches the MLG bands (right inset, Fig. 3e) a strong feature appears in dI^2/dV^2 . Consequently, the two TBG Dirac points trace out two copies of the MLG bands, displaced by $\Delta\theta = 2.7^\circ$. We use the MLG Dirac bands to calibrate the energy shift at each V_b (Supplementary Information

Section 8). The extracted flat bands energy dispersion (dots, Fig. 3g) compares well with the Bistrizer–MacDonald model (lines). Furthermore, the measurements show the layer polarization of individual wavefunctions at different momenta: visibly, the dI^2/dV^2 amplitudes are strong when the MLG lattice is rotationally aligned with the top TBG layer ($\theta = +1.35^\circ$) but are substantially weaker when it is aligned with the bottom TBG layer ($\theta = -1.35^\circ$), showing that the experiment probes the weight of the wavefunction on the top layer. Plotting the magnitude of I along the traced flat band features (blue, Fig. 3f) between K_{top} and K_{bot} (Fig. 3h, dots) indeed shows nice correspondence with the k -dependent layer polarization calculated from the Bistrizer–MacDonald model (line).

Imaging TBG energy bands under pressure

We end by showing the ability to tune the local pressure in the QTM junction using AFM applied forces (Fig 4a, inset), thereby tuning the interlayer tunnelling^{30–34}. Figure 4a–c plots the measured dI^2/dV^2 versus θ and V_b , for the junction of Fig. 3, with pressures $p = 0.01, 0.4$ and 0.68 GPa (twisting is performed with the pressure constantly on). Visibly, with increasing p , the flat bands shrink towards zero V_b , whereas the remote bands get further away (Supplementary Video 1). Tracing the bands over a larger sequence of pressures (Fig. 4d) shows the opposite dependence of the flat and remote bands on pressure (arrows), reflecting a band anticrossing that increases with p , as expected from increased interlayer tunnelling. Converting V_b to the energy shift using the simultaneously measured MLG bands (Fig. 4e) shows that flat bands width shrinks linearly with p , reaching a 17% reduction at $p = 0.68$ GPa. This compares reasonably to the 6–14% reduction predicted theoretically^{30,33,34}. A naïve linear extrapolation of our measurements would suggest that 2.7° TBG bands would become fully flat at $p \cong 4$ GPa, well within the pressures achievable by AFM without damaging graphene^{35,36}.

The QTM opens the way for two independent research directions: first, it provides a new approach for creating highly controllable new interfaces between a large variety of quantum materials. Specifically, it enables continuous control of a critical parameter of these interfaces: their twist angle. The technique should be applicable generally to a plethora of layered conductors^{4,37–41}, semiconductors^{41,42} and superconductors^{43–45}. Second, it is a new scanning microscope that has direct access to the energy-momentum dispersion of electronic systems. As such, it may probe the dispersion of any excitation, charged or neutral, if it is excited by a tunnelling electron. Measurements can be performed at large magnetic fields, with variable carrier density and electrical displacement fields controlled by local gates, and with a continuously tunable pressure. The lateral scanning capabilities of the QTM will further enable spatially scanned momentum-resolved measurements within electronic devices with a high spatial resolution (roughly 100 nm). Given the relative simplicity of the technique and its powerful capabilities, we expect the QTM to become a valuable new tool in the arsenal of experimental condensed matter physics.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-022-05685-y>.

1. Binnig, G. & Rohrer, H. Scanning tunneling microscopy. *Surf. Sci.* **152–153**, 17–26 (1985).
2. Bian, K. et al. Scanning probe microscopy. *Nat. Rev. Methods Primers* **1**, 36 (2021).
3. Bistrizer, R. & MacDonald, A. H. Moiré bands in twisted double-layer graphene. *Proc. Natl Acad. Sci. USA* **108**, 12233–12237 (2011).
4. Cao, Y. et al. Unconventional superconductivity in magic-angle graphene superlattices. *Nature* **556**, 43–50 (2018).
5. Cao, Y. et al. Correlated insulator behaviour at half-filling in magic-angle graphene superlattices. *Nature* **556**, 80–84 (2018).

6. Eisenstein, J. P., Gramila, T. J., Pfeiffer, L. N. & West, K. W. Probing a two-dimensional Fermi surface by tunneling. *Phys. Rev. B* **44**, 6511–6514 (1991).
7. Murphy, S. Q., Eisenstein, J. P., Pfeiffer, L. N. & West, K. W. Lifetime of two-dimensional electrons measured by tunneling spectroscopy. *Phys. Rev. B* **52**, 14825–14828 (1995).
8. Auslaender, O. M. et al. Tunneling spectroscopy of the elementary excitations in a one-dimensional wire. *Science* **295**, 825–828 (2002).
9. Britnell, L. et al. Resonant tunnelling and negative differential conductance in graphene transistors. *Nat. Commun.* **4**, 1794 (2013).
10. Mishchenko, A. et al. Twist-controlled resonant tunnelling in graphene/boron nitride/graphene heterostructures. *Nat. Nanotechnol.* **9**, 808–813 (2014).
11. Fallahazad, B. et al. Gate-tunable resonant tunneling in double bilayer graphene heterostructures. *Nano Lett.* **15**, 428–433 (2015).
12. Wallbank, J. R. et al. Tuning the valley and chiral quantum state of Dirac electrons in van der Waals heterostructures. *Science* **353**, 575–579 (2016).
13. Jang, J. et al. Full momentum- and energy-resolved spectral function of a 2D electronic system. *Science* **358**, 901–906 (2017).
14. Prasad, N. et al. Quantum lifetime spectroscopy and magnetotunneling in double bilayer graphene heterostructures. *Phys. Rev. Lett.* **127**, 117701 (2021).
15. Lin, K. et al. Emergence of interlayer coherence in twist-controlled graphene double layers. *Phys. Rev. Lett.* **129**, 187701 (2022).
16. Seo, Y. et al. Subband-resolved momentum-conserved resonant tunneling in monolayer graphene/h-BN/ABA-trilayer graphene small-twist-angle tunneling device. *Appl. Phys. Lett.* **120**, 083102 (2022).
17. Koren, E. et al. Coherent commensurate electronic states at the interface between misoriented graphene layers. *Nat. Nanotechnol.* **11**, 752–757 (2016).
18. Chari, T., Ribeiro-Palau, R., Dean, C. R. & Shepard, K. Resistivity of rotated graphite-graphene contacts. *Nano Lett.* **16**, 4477–4482 (2016).
19. Ribeiro-Palau, R. et al. Twistable electronics with dynamically rotatable heterostructures. *Science* **361**, 690–693 (2018).
20. Yang, Y. et al. In situ manipulation of van der Waals heterostructures for twistronics. *Sci. Adv.* **6**, eabd3655 (2020).
21. Hu, C. et al. In-situ twistable bilayer graphene. *Sci. Rep.* **12**, 204 (2022).
22. Frisenda, R. et al. Recent progress in the assembly of nanodevices and van der Waals heterostructures by deterministic placement of 2D materials. *Chem. Soc. Rev.* **47**, 53–68 (2018).
23. Bistrizer, R. & MacDonald, A. H. Transport between twisted graphene layers. *Phys. Rev. B* **81**, 245412 (2010).
24. Mele, E. J. Commensuration and interlayer coherence in twisted bilayer graphene. *Phys. Rev. B* **81**, 161405 (2010).
25. Damascelli, A., Hussain, Z. & Shen, Z.-X. Angle-resolved photoemission studies of the cuprate superconductors. *Rev. Mod. Phys.* **75**, 473–541 (2003).
26. Zhang, H. et al. Angle-resolved photoemission spectroscopy. *Nat. Rev. Methods Primers* **2**, 54 (2022).
27. Feenstra, R. M., Jena, D. & Gu, G. Single-particle tunneling in doped graphene-insulator-graphene junctions. *J. Appl. Phys.* **111**, 043711 (2012).
28. Elias, D. C. et al. Dirac cones reshaped by interaction effects in suspended graphene. *Nat. Phys.* **7**, 701–704 (2011).
29. Lopes dos Santos, J. M. B., Peres, N. M. R. & Castro Neto, A. H. Continuum model of the twisted graphene bilayer. *Phys. Rev. B* **86**, 155449 (2012).
30. Carr, S., Fang, S., Jarillo-Herrero, P. & Kaxiras, E. Pressure dependence of the magic twist angle in graphene superlattices. *Phys. Rev. B* **98**, 085144 (2018).
31. Yankowitz, M. et al. Dynamic band-structure tuning of graphene moiré superlattices with pressure. *Nature* **557**, 404–408 (2018).
32. Yankowitz, M. et al. Tuning superconductivity in twisted bilayer graphene. *Science* **363**, 1059–1064 (2019).
33. Chittari, B. L., Leconte, N., Javvaji, S. & Jung, J. Pressure induced compression of flatbands in twisted bilayer graphene. *Electron. Struct.* **1**, 015001 (2018).
34. Chebrolov, N. R., Chittari, B. L. & Jung, J. Flat bands in twisted double bilayer graphene. *Phys. Rev. B* **99**, 235417 (2019).
35. Lee, C., Wei, X., Kysar, J. W. & Hone, J. Measurement of the elastic properties and intrinsic strength of monolayer graphene. *Science* **321**, 385–388 (2008).
36. Ares, P. et al. Tunable graphene electronics with local ultrahigh pressure. *Adv. Funct. Mater.* **29**, 1806715 (2019).
37. Chen, G. et al. Evidence of a gate-tunable Mott insulator in a trilayer graphene moiré superlattice. *Nat. Phys.* **15**, 237–241 (2019).
38. Zhou, H., Xie, T., Taniguchi, T., Watanabe, K. & Young, A. F. Superconductivity in rhombohedral trilayer graphene. *Nature* **598**, 434–438 (2021).
39. Zhou, H. et al. Half- and quarter-metals in rhombohedral trilayer graphene. *Nature* **598**, 429–433 (2021).
40. Shi, Y. et al. Electronic phase separation in multilayer rhombohedral graphite. *Nature* **584**, 210–214 (2020).
41. Kennes, D. M. et al. Moiré heterostructures as a condensed-matter quantum simulator. *Nat. Phys.* **17**, 155–163 (2021).
42. Mak, K. F. & Shan, J. Semiconductor moiré materials. *Nat. Nanotechnol.* **17**, 686–695 (2022).
43. Yabuki, N. et al. Supercurrent in van der Waals Josephson junction. *Nat. Commun.* **7**, 10616 (2016).
44. Farrar, L. S. et al. Superconducting quantum interference in twisted van der Waals heterostructures. *Nano Lett.* **21**, 6725–6731 (2021).
45. Zhao, S. Y. F. et al. Emergent interfacial superconductivity between twisted cuprate superconductors. Preprint at <https://arxiv.org/abs/2108.13455> (2021).

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Fabrication of vdW device-on-tip

We start with commercial tipless AFM cantilevers with a nominal spring constant of $K = 40 \text{ N m}^{-1}$, length of $L = 232 \text{ }\mu\text{m}$, width of $W = 37 \text{ }\mu\text{m}$ and thickness of $T = 7 \text{ }\mu\text{m}$ (NANOSENSORS). We coat the bottom side of the cantilevers with 4/100 nm Cr/Au using an e-beam evaporator. This coating serves as the electrical connection for the active vdW layer and the buried graphite, such that they are at the same electrical potential. Next, we deposit a 1.2–1.6 μm tall platinum (Pt) pyramid with a 2 μm square base using FIB deposition (Supplementary Information Section 1). The nominal size of the pyramids' apex is roughly 50 nm. To assemble the vdW device-on-tip, we use the polymer membrane transfer technique. Briefly, we spin-coat polypropylene carbonate (PPC) onto a Si/(90 nm) SiO_2 wafer, followed by exfoliating graphene/hBN crystals. We then cut a window into a piece of adhesive tape and use it to pick up the PPC film with the target crystal. The crystal is then aligned and transferred on top of the platinum pyramid, followed by melting (roughly 120 °C) and washing off (acetone) the PPC film to leave behind the vdW crystal. We assemble the vdW heterostructure by successive transferring and cleaning of each layer. For example, we start by transferring a graphite flake, 20–40 nm thick, on top of the Pt pyramid, followed by an insulating hBN flake, 20–40 nm. These two layers act as an atomically flat mechanical support for the active vdW layer (MLG), which is transferred at the last step. The thicker underlayers also define the shape of the plateau obtained at the tip apex. By varying the thickness of these layers, we can obtain plateaus of variable sizes in the range of 50 nm to 1 μm .

Fabrication of vdW device on flat substrate

Our devices sitting on a flat substrate are assembled using the standard dry transfer technique. Briefly, a PPC coated polydimethylsiloxane (PDMS) is used to pick up the hBN, MLG (or TBG created using the laser cut-and-stack method) and WSe_2 layers. We then flip the stacks by depositing them onto a poly(methyl methacrylate)-coated PDMS, followed by placing them on the bottom graphite gate. The final heterostructure is transferred on a roughly 100 μm wide and 80 μm tall Si/ SiO_2 pillar, with pre-patterned gold contacts reaching close to its edge (Supplementary Information Section 2). After assembly, the heterostructures are cleaned by scanning them with an AFM in contact mode to remove polymer residues.

Measurements

Our QTM setup comprises two parts: (1) a custom-built rotating stage housing the vdW devices on flat substrate and (2) a commercial AFM (Bruker Icon) for performing the AFM operations on the vdW device-on-tip (force feedback and lateral positioning). The custom-built rotating stage comprises (from bottom to top): a goniometer,

piezoelectric rotator (attocube), X and Y piezoelectric nanopositioners (attocube) and a sample holder with several electrical contacts (Supplementary Information Section 3). The piezoelectric rotator has an optical encoder allowing for closed-loop angular control with roughly 1 milli-degree precision. The two linear nanopositioners, mounted on top of the piezoelectric rotator, position the device's point-of-interest in the centre of rotation. This custom stage is attached using a rigid frame to an existing AFM stage. The adapter allows us to use the integrated optics and all standard AFM operations. The AFM enables us to bring the two vdW surfaces (in the tip and the sample) into contact. Additionally, we use the active force feedback to maintain a constant force during rotation and measurement with a sub-nano-Newton precision. With this QTM setup, we use two modalities for the rotation measurements. (1) Continuous contact: here, we bring the tip and sample together and rotate while maintaining the two surfaces in contact throughout the entire experiment. (2) 'Woodpecker': here, for every point in the measurement as a function of rotation angle, we bring the tip into contact with the sample, perform the measurement, disengage, then rotate by a small angle and then repeat this sequence to get the full measurement. We note that we obtain almost identical results from both modalities, and both are highly reproducible.

Voltages are sourced using a home-built digital-to-analogue converter (DAC) array and measured using a home-built, software-based audio-frequency lock-in amplifier consisting of 1 μV accurate d.c. + a.c. sources and a Femto DPLCA-200 current amplifier and NI-9239 sampler.

Data availability

The data shown in this paper are provided with this paper. Additional data that support the plots and other analysis in this work are available from the corresponding author upon request.

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Author contributions A.I., J.B., J.X. and S.I. designed the experiment. A.I., J.B. and J.X. built the setup, fabricated the devices and performed the experiments. A.I., J.B., J.X. and S.I. analysed the data. J.X., B.Y., Y.O., A.S. and E.B. wrote the theoretical models. K.W. and T.T. supplied the hBN crystals. A.I., J.B., J.X., A.S., E.B. and S.I. wrote the manuscript with input from other authors.

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Additional information

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